TINA REZVANIAN

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SUMMARY

- Data Scientist/Machine Learning Expert with 7.5 years of proficiency in vision and language generative AI, personalization, and recommendation leveraging both structured and unstructured data
- 1.5 years in technical leadership, spearheading Proof of Concepts and product model development
- Proficient in advanced ML/DS modeling, encompassing supervised, unsupervised learning, and GenAI
- Adept at managing end-to-end pipelines and deploying models effectively

EDUCATION

Harvard Extension School — Online
Graduate Degree, Deep Learning — 2019

Northeastern University — Boston, MA
Ph.D., Industrial Engineering and Operations Research | Minor in Mathematics — 2013 - 2019

Northeastern Illinois University — Chicago, IL
MBA, Master of Business Administration — 2011 - 2013

University of Science and Technology of Mazandaran — Babol, Iran
B.Sc., Industrial Systems Engineering — 2006 - 2010

SKILLS

GenAI/ ML/DS Vision & Language Transformers, Variational Auto-Encoders, Stable Diffusion, Different Attention Mechanisms, Convolutional NN, Deep Feedforward Networks, Dimension Reduction, KV caching, Normalization & Regularization Methods, Gradient-based & Tree-based Methods, Supervised, Unsupervised, Ensemble, Support Vector Machines, Causal Models, Clustering

Programming Python, TensorFlow, PyTorch, Spacy, R, MATLAB, Flask, AWS, CPLEX, SQL, Hadoop

Visualization Matplotlib, Plotly, Seaborn, Pyvis, ggplot2, Gephi, Tableau, Power BI

WORK EXPERIENCE

Adobe — Sep 2019 - Present
Data Scientist, Machine Learning, Firefly — *San Jose, CA*

- LLM & VLM on Firefly: Finetuned pre-trained LLM & VLMs to improve Firefly's performance
  - Enriched training captions, tags, and images through the generation of more advanced captions, GenAI images, and embeddings of different sizes
  - Inferred from, trained, or finetuned GenAI models including Llama 2, Stable Diffusion, LDM VAE
  - Evaluated model performances, investigated the impact of modifications in output images and quantified their impact on user satisfaction and image technical quality
- AB Testing on Firefly: Evaluated the impact of data enrichment strategies on Firefly performance, developed a framework for rapid experimentation, and automated training/retraining/fine-tuning
  - Conducted experiments with curated datasets to address model limitations, integrating advanced architectural techniques in diffusion UNET, and employing multi-representational embeddings
  - Enhanced model inference, output comparison, and metric evaluation, tracked progress by integration to Weights & Biases and Streamlit for human-centric app evaluations
  - Automated connectivity to RunAI, AWS, and Artifactory; streamlined job submission, user input, components configurations, checkpoints, docker image, GPUs; facilitated seamless transfer of checkpoints, input and output data

- Data Enrichment Model Gap Analysis: Improved image generation through identification and rectification of content gaps in training images and false representations in feature embeddings from models handling text presence, aesthetic scoring, face/human detection, and caption generation
  - Uncovered and remedied gaps in subjects, contexts, and actions within training images and captions
  - Identified and resolved inaccuracies in image representations derived from data enrichment models
  - Conducted comprehensive NLP analysis of caption quality, focusing on model bias in part of speech selection, content categorization, and image/caption alignment using CLIP score
- Multi-Channel Tool Attribution: Boosted retention and conversion rates by offline learning of personalized next best action and time of sending in-app messages during the user journey
  - Translated business needs to measurable objectives through collaboration with cross-functional teams
  - Built data pipelines, designed a multi-head attention model to estimate the impact features, time-of-use, and critical workflows per user on engagement
  - Utilized predicted attrition probabilities for high-credit time steps to optimize targeting and campaign
- Super Attributes: Characterized users and revamped user segmentation using unsupervised methods to optimize retention-focused workflows and personalize onboarding, in-app, and email communications
  - Re-purposed β-variational auto-encoder to automatically extract meaningful attributes from hidden features with minimum KL divergence from the standard normal distribution
  - Clustered users through Gaussian Mixture Models using normally distributed learned attributes
  - Generated similar sequences to that of retained users from decoder β-VAE model within each cluster
- FontMART: Ranked fonts that maximize reading speed for individual users through pair-wise learning
  - Gathered and evaluated from MTurk, web apps, and Hadoop to ensure data quality and readiness
  - Implemented LamdaMart model to directly learn NDCG based on reader and font characteristics
  - Identified globally optimal fonts along with fastest fonts over different user segments
  - Determined influential font characteristics on reading spread across different age groups
  - Compared preferred, best-performing, and predicted fonts, and accuracy conditions
- RealTime Recommender: Developed a real-time recommender on the device, leveraging reinforcement learning (RL) to train the recommender through interaction with a generative AI acting as a user agent
  - Trained a generative adversarial network to create a generalized user model from sequential data
  - Employed deep Q-learning to build a general recommender using the previously trained user model
  - Personalized the recommender by fine-tuning both the user model and recommender for specific users
- CausalTIM: Identified feature's impact on conversion and engagement by simulating a randomized control trial using observed data instead of running experiments
  - Employed propensity score matching to assign feature-users to non-users depicting similar behaviors
  - Implemented parallelization strategies in running separate models predicting features' causal effects

Harvard Extension School Nov 2019 - May 2020

- Face Recognition: Built face recognition models using convolutional neural networks to match multiple extracted faces in images to known names or unknown identity in CASIA-WebFace dataset
  - Re-purposed FaceNet, DeepFace, VGGFace, and ResNet models to identify new identities using feature extraction, transfer learning, fine-tuning, data augmentation, and regularization strategies

Insight Data Science Jan 2019 - Feb 2019

- PickaBasket: Built a web application accessible at pickabasket.com using Python, Flask, and AWS for offering bestseller bundles for online grocery stores
  - Identified patterns in customers' purchase behavior as probabilistic rules on bestseller combinations

Wheels UP Private Jet Company May 2018 - Aug 2018
Data Science Intern *New York City, NY*

- Forecasting Daily Flight Demand and Allocating Aircraft to Regions to Minimize Empty Flights
  - . Clustered flight movements by designing regional/connectivity based similarity metrics to use in community detection methods such as agglomerative nesting and edge-betweenness algorithms

Northeastern University Dec 2014 - Sep 2019
Research Assistant *Boston, MA*

- Forecasted Emergency Response (ER) Demand for Refugees as a Zero-Inflated Skewed Variables at United Nations High Commissioner for Refugees (UNHCR supported)
  - . Developed a two-stage heuristic that outperforms Tweedie and Poisson zero-inflated models
  - . Predicted the amount of ER across all locations using trees and calculated the expected ER demand
- Developed a Matching Algorithm on Short Preference Lists at World Food Programme (NSF supported)
  - . Introduced the concept of negotiations in the Top Trading Cycle algorithm in multi-periodic markets
  - . Guaranteed a minimum match quality by the adoption of α-stable matches defined over all periods
  - . Increased total welfare by 40% and total matched pairs by 80%, and reduced turnover rate by 83%
  - . Wrote and awarded 400K NSF grant, delivered a decision support tool using Python and Xlwings
- Integrated Emergency and Ongoing Supply Chains in a Multi-Criteria Framework (UNHCR supported)
  - . Minimized total costs and response time in constrained two-stage stochastic mixed integer model
  - . Reduced total cost by 31%, and improved response time by 18%
  - . Developed an open source decision support tool for UNHCR supply chains group GLPK and Python

Healthcare Systems Engineering Institute (HSYE) Sep 2013 - Dec 2014
Northeastern University, Research Assistant *Boston, MA*

- Scheduled Treatments and Nurses using Column Generation Algorithm at Moffit Cancer Center
- Optimized Curbside Threshold at Massachusetts General Hospital (MGH) Neurology Department

PUBLICATIONS

. Beier, Sofie, et al. ”Readability research: An interdisciplinary approach.” Foundations and Trends® in Human–Computer Interaction 16.4 (2022): 214-324.

. Cai,T. , Wallace,S., **Rezvanian,T.**(2022, June). Personalized Font Recommendations:Combining ML and Typographic Guidelines to Optimize Readability. In Designing Interactive Systems Conference(pp.1-25)

. Jahre,M.,Kembro,J., **Rezvanian,T.**, Ergun.O, Hapnes(2016)*Integrating supply chains for emergencies and ongoing operations in UNHCR*, Journal of Operations Management,45,57-72.

. Jahre, M., Kembro, J., **Rezvanian, T.**, Ergun, O., Haapnes, S.,& Hultberg, M. *Scenario building in UNHCR-Predicting future demand in refugee crises.* In Proceedings of the 5th World Production and Operations Management Conference, POM Havana 2016

. **Rezvanian, T.**, Ergun, O., Jahre, M., Kembro, J., (2020) *Forecasting Emergency Response Demand for Refugees as a Zero-Inflated Skewed variables: Evidence from UNHCR*

. **Rezvanian, T.**, Ergun, O., *Two-sided Stable Matching with Negotiations*, Integrating Data-Driven Forecasting and Large-Scale Optimization to Improve Humanitarian Response Planning and Preparedness. Chapter 1, Diss. Northeastern University,

. **Rezvanian, T.**, Ergun, O., *Deferred Acceptance Algorithms and Aftermarkets for Matching with Short Preference Lists at World Food Programme*, Integrating Data-Driven Forecasting and Large-Scale Optimization to Improve Humanitarian Response Planning and Preparedness. Chapter 1, Diss. Northeastern University, 2019.